Updated Screening-Level Coefficients for Dairy Manure Management Systems

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June 3rd, 2026

Introduction

This report documents the development of updated screening-level coefficients for dairy manure management systems. The coefficients are intended for use in a Manure Sub-Surface Drip Irrigation (MSDI) economic screening model that tracks solids, liquids, and nitrogen through collection, separation, storage, digestion, and downstream irrigation systems. The updated values replace or refine an existing set of placeholder coefficients for solids retention, liquid retention, and total nitrogen retention in the liquid stream across a set of manure management practices.

The purpose of this update was not to create a site-specific manure mass balance for each dairy configuration. Manure management conditions vary substantially across the United States because of differences in housing, bedding, manure collection, flush water use, separator operation, storage residence time, climate, and downstream land application practices [1]. Instead, this analysis develops transparent, repeatable, and defensible screening-level coefficients that can be used to capture relative performance of a broad set of dairy manure management scenarios and practices. Where possible, coefficients were developed from values reported in the literature, reputable technical references, and documented manure management studies. Preference was given to values with transparent methods, grounding in measured or modeled data, and clear relevance to dairy manure systems. Where direct values were unavailable, coefficients were retained or adjusted using explicit engineering assumptions.

The coefficients are defined as downstream retention multipliers. A solids retention coefficient of 0.70 means that 70% of incoming solids remain in the downstream manure stream after the process. A liquid retention coefficient of 0.90 means that 90% of incoming liquid remains in the downstream liquid stream after the process. A TN liquid multiplier represents the fraction of liquid-phase nitrogen retained in the liquid stream after accounting for nitrogen losses or dilution, depending on the process type.

1. Solids Retention Coefficients

1.1 Collection Systems

Collection and conveyance systems were treated as screening-level transfer processes rather than treatment or degradation systems. These systems include gutters, chain and paddle systems, scraping systems, vacuum collection, flush systems, and gravity flow or slatted floor systems. Because these systems primarily move manure from housing areas to downstream handling or storage, solids coefficients were interpreted as the fraction of manure solids physically conveyed downstream.

For non-flush collection systems, updated coefficients were based on expected physical collection performance [1], [2], [3]. Guided mechanical systems, such as gutter or chain and paddle systems, were assumed to have high solids transfer because manure follows a defined conveyance path. Tractor or skid scraping was treated as a bulk-solids collection method that captures most manure solids but leaves residual manure on the floor. Hydraulic or mechanical scraping was assigned a higher solids transfer coefficient than tractor/skid scraping because these systems operate more consistently and repeatedly. Vacuum systems were represented as liquid/slurry collection systems with moderate solids capture, limited by heavier settled solids and residual manure left on floor surfaces. Gravity flow and slatted floor systems were assumed to pass liquid and smaller particles readily while allowing some coarse solids to bridge, settle, or accumulate before downstream transfer.

Flush collection systems were handled differently. High-volume and automated/precision flush systems were assumed to transfer essentially all mobilized manure solids to downstream liquid handling systems. Therefore, the solids retention coefficient was kept at 1.00 for both flush categories.

1.2 Separation Systems

Fiber separation solids coefficients were based on reported volatile solids removal efficiencies and treated as proxies for total solids removal. This approximation assumes that the removed solids have approximately the same VS/TS ratio as the incoming manure for purposes of estimating the fraction of total solids removed. Under that assumption, the fraction of VS removed can be applied directly as the fraction of TS removed.

The general relationship was:

$$S_{out} = 1 - f_{TS,rem} \qquad \text{(Eq. 1)}$$

where S_{out} is the solids retention coefficient in the downstream liquid effluent and $f_{TS,rem}$ is the fraction of incoming TS removed. For fiber separation, $f_{TS,rem}$ was approximated using reported VS removal efficiencies.

Values for slope screen, screw press, rotary drum, vibratory/shaker screen, centrifuge, and settling basin systems were based on Greene et al. [1], Williams et al. [4], CARB GREET [5], Chastain and Henry [6], Møller et al. [7], and Zhang et al. [8]. Because solids and liquid retention can vary substantially with influent solids loading, two additional categories were recommended: screw press under high-solids loading and low-solids loading, and centrifuge under high-solids loading and low-solids loading. These categories allow the model to better represent differences between flush-type manure streams and higher-solids scrape/vacuum manure streams.

1.3 Storage and Digestion Systems

Solids coefficients for manure storage systems were based on a reduced-order conversion from VS destruction to TS retention. For lagoon and liquid/slurry systems, IPCC methane conversion factors were used as screening-level proxies for VS conversion/destruction. Because MCF values are expressed on a volatile-solids basis and total solids include fixed solids that are not biologically degraded, MCF values were converted to TS retention coefficients using scenario-specific VS/TS ratios.

The general relationship was:

$$TS_{ret} = 1 - (MCF \times VS/TS) \qquad \text{(Eq. 2)}$$

where TS_{ret} is the total solids retention coefficient, MCF is the methane conversion factor used as a proxy for VS conversion, and VS/TS is the volatile-solids fraction of total solids for the relevant manure stream.

For primary lagoons, the updated coefficient used the average of the IPCC temperate uncovered anaerobic lagoon MCFs. The average lagoon MCF was 0.69 [2]. A lagoon-water VS/TS ratio of 0.50 was used, consistent with reported lagoon water values of approximately 48 to 52% VS/TS in Williams et al. [4]. The resulting coefficient was:

$$TS_{ret,primary} = 1 - (0.69 \times 0.50) = 0.65 \quad \text{(Eq. 3)}$$

For secondary lagoons and effluent ponds, the same lagoon basis was used, but an explicit residual-degradability adjustment was applied because the primary lagoon has already removed or degraded a portion of the readily degradable and settleable organic matter. The secondary lagoon effective MCF was assumed to be 75% of the primary lagoon MCF:

$$MCF_{secondary} = 0.75 \times MCF_{primary} = 0.5175 \quad \text{(Eq. 4)}$$

$$TS_{ret,secondary} = 1 - (0.5175 \times 0.50) = 0.74 \quad \text{(Eq. 5)}$$

This approach retains the lagoon classification for secondary lagoons and effluent ponds while recognizing lower incremental VS destruction after primary lagoon treatment.

For covered lagoon digesters, a solids retention coefficient of 0.60 was selected. This assumes moderately enhanced VS destruction relative to an uncovered lagoon because the cover reduces evaporative cooling, limits oxygen intrusion, improves anaerobic stability, and enables intentional methane capture, while still recognizing incomplete degradation of total solids. The coefficient remains screening-level because the cover primarily affects gas capture and operating conditions rather than creating a fundamentally different solids-destruction mechanism.

For engineered digesters, including plug-flow, complete-mix, and fixed-film digesters, a common solids retention coefficient was formulated based on Greene et al. [1]. Greene et al. estimated a degradable VS fraction of 51.7% and assumed engineered complete-mix anaerobic digestion performance based on methane production efficiency. A digester conversion efficiency of 90% was used as a screening proxy for engineered dairy anaerobic digestion, and a collected manure VS/TS ratio of 0.70 was used [4]. The resulting coefficient was:

$$TS_{ret,digesters} = 1 - (0.517 \times 0.90 \times 0.70) = 0.67 \quad \text{(Eq. 6)}$$

This value was applied to plug-flow, complete-mix, and fixed-film digesters. Other or specialized digesters were assigned a lower retention coefficient of 0.50 to represent higher solids destruction for specialized or advanced digestion systems.

Liquid/slurry storage values were included because liquid/slurry storage is both a common standalone manure management pathway and a common downstream digestate storage option after engineered anaerobic digestion [1]. Including separate liquid/slurry categories also improves model flexibility because users may need to represent digestate storage after plug-flow, complete-mix, fixed-film, or other enclosed digesters without routing that material through a lagoon or effluent pond. Two liquid/slurry categories were included to distinguish systems with a natural crust cover from uncovered systems, since crust formation can materially affect methane conversion, evaporation, and nitrogen volatilization [2].

For liquid/slurry storage, the average of the IPCC temperate six-month liquid/slurry MCFs was used. The average MCF was 0.313 [2]. For liquid/slurry systems with a natural crust cover, IPCC guidance allows a 40% reduction in MCF when a thick, dry natural crust is present. Therefore, the effective MCF for liquid/slurry with natural cover was:

$$MCF_{natural\ cover} = 0.313 \times 0.6 = 0.188 \qquad \text{(Eq. 7)}$$

A higher-solids scrape/vacuum VS/TS ratio of 0.70 was used for liquid/slurry storage because these streams are less diluted and less biologically stabilized than lagoon water but lower in VS/TS than as-excreted manure due to bedding, sand, soil, and inorganic solids contamination [4].The resulting coefficients were:

$$TS_{ret,natural\ cover} = 1 - (0.188 \times 0.70) = 0.87 \qquad \text{(Eq. 8)}$$

$$TS_{ret,uncovered} = 1 - (0.313 \times 0.70) = 0.78 \qquad \text{(Eq. 9)}$$

All updated solids retention coefficients are provided in Table 1.

Table 1. Updated screening-level solids retention coefficients for dairy manure management systems.

Manure Management System	Solids Change (Prev.)	Solids Change (Updated)	Methods/Justification
Removal-Gutter / Chain and Paddle	0.9	0.9	Unchanged. Treated as a guided mechanical conveyance system with high solids capture and limited opportunity for solids loss before downstream handling.
Removal-Scrape (Tractor / Skid)	0.8	0.8	Unchanged. Treated as a bulk-solids collection method with moderate residual manure left on the floor after scraping.
Removal-Scrape (Hydraulic / Mechanical)	0.7	0.85	Updated. Mechanical/hydraulic scrape systems should be at least as effective as tractor/skid scraping for repeatable solids removal; coefficient reflects improved consistency relative to mobile equipment scraping.
Removal-Vacuum Truck	0.6	0.7	Updated. Vacuum systems are better represented as liquid/slurry collection systems with moderate solids capture, limited by heavier settled solids and floor residuals.
Removal-Flush (High Volume)	1	1	Unchanged. Assumes high-volume flush transfers essentially all mobilized manure solids to the downstream liquid handling system.
Removal-Flush (Automated/Precision)	1	1	Unchanged. Assumes automated/precision flush transfers essentially all mobilized manure solids downstream.
Removal-Gravity Flow / Slatted Floor	0.7	0.7	Unchanged. Liquid and smaller particles pass readily, but coarse solids can bridge, settle, or accumulate before downstream transfer.
Storage-Primary Lagoon	0.6	0.65	Updated using IPCC uncovered anaerobic lagoon MCFs as a screening-level proxy for VS conversion, converted to TS retention using lagoon-water VS/TS=0.50: 1−(0.69×0.50) = 0.65
Storage-Secondary Lagoon/Effluent Pond	0.8	0.74	Same lagoon basis, but with an explicit residual-degradability adjustment: secondary lagoon effective MCF = 75% of primary lagoon MCF, giving 1−(0.5175×0.50) = 0.74
Liquid/Slurry (Natural Cover) - Added	**-**	**0.87**	**Based on IPCC 6-month liquid/slurry MCF for temperate systems with a 40% reduction for thick natural crust cover: (MCF = 0.313×0.60 = 0.188). Converted to TS retention using higher-solids scrape/vacuum (VS/TS = 0.70): 1 - (0.188×0.70) = 0.87**
Liquid/Slurry (Uncovered) - Added	**-**	**0.78**	**Based on IPCC 6-month liquid/slurry MCF for temperate systems: (MCF = 0.313). Converted to TS retention using higher-solids scrape/vacuum (VS/TS = 0.70): 1 - (0.313×0.70) = 0.78**
Storage-Digester (Covered lagoon)	0.5	0.6	Covered lagoon digester (0.60) Assumes moderately enhanced VS destruction relative to an uncovered lagoon because the cover reduces evaporative cooling, limits oxygen intrusion, improves anaerobic stability, and enables intentional methane capture, while still recognizing incomplete degradation of total solids.
Storage-Digester (Plug-flow)	0.5	0.67	Assumed same as Complete-Mix

Storage-Digester (Complete-mix)	0.6	0.67	Updated using the Greene et al. degradable VS fraction ($f_{deg,VS} = 0.517$), assumed engineered digester performance as 90% of maximum methane potential and collected manure VS/TS = 0.70: 1 - (0.517×0.90×0.70) = 0.67
Storage-Digester(Fixed-film)	0.6	0.67	Assumed same as Complete-Mix
Storage-Digester (Other/Specialized)	0.4	0.50	Assumes a substantial increase in VS destruction relative to Complete-Mix (due to specialized or enhanced design and/or construction)
Fiber Separation-Slope Screen	0.4	0.70	Based on VS removal from Williams et al. 2020 (Table 2-5)
Fiber Separation-Screw Press (High Solids)	0.35	0.50	Based on VS removal from Williams et al. 2020 (Table 2-5)
Fiber Separation-Screw Press (Low Solids) - Added	**-**	**0.75**	**Based on VS removal from Williams et al. 2020 (Table 2-5)**
Fiber Separation-Rotary Drum	0.5	0.75	Based on VS removal from CARB GREET
Fiber Separation-Vibratory / Shaker Screen	0.6	0.85	Based on VS removal from CARB GREET
Fiber Separation-Centrifuge (High Solids)	0.2	0.30	Based on VS removal reported in Zhang et al. (2022) and Moller et al. (2002)
Fiber Separation-Centrifuge (Low Solids) - Added	**-**	**0.50**	**Based on VS removal from CARB GREET**
Fiber Separation-Settling Basin	0.6	0.50	Based on VS removal Chastain & Henry 2015
Solids Removal-Stacking, Bedding, Etc.	0.1	0.1	Unchanged
Field / Off-Site	1	1	Unchanged
Irrigation-Sand Pots / Sand Media Filters	0.15	0.15	Unchanged
N/A (No Treatment)	1	1	Unchanged

2. Liquid Retention Coefficients

2.1 Collection Systems

Collection-system liquid coefficients were treated as physical transfer efficiencies. Guided conveyance systems were assumed to retain most liquid manure in the downstream stream, with only minor residual liquid left behind. Scrape systems were assumed to recover less liquid than solids because scraping captures bulk manure but leaves free liquid and thin liquid films on housing surfaces. Vacuum systems were assigned higher liquid retention because they are designed to recover liquid/slurry manure. Gravity flow and slatted floor systems were assumed to retain most liquid in the downstream manure pathway.

Flush systems were treated as dilution systems rather than conventional liquid-retention systems. High-volume flush was assigned a liquid multiplier of 20, and automated/precision flush was assigned a liquid multiplier of 8. These coefficients represent dilution from flush water rather than physical liquid retention. Flush water was assumed to transfer manure downstream while increasing the liquid volume of the manure stream.

2.2 Separation Systems

Liquid retention through fiber separation was estimated using a solids-liquid mass balance on removed total solids and entrained water in the separated solids fraction. VS removal efficiencies were used as a proxy for TS removal, and the wet mass of separated solids was calculated from the assumed TS content of the separated solids cake. The resulting liquid coefficient represents the fraction of incoming liquid mass remaining in the downstream liquid stream after accounting for water retained in removed solids. This approach assumes that liquid loss occurs primarily through water entrained in separated solids rather than through an independent liquid removal mechanism.

The relationship was:

$$L_{out} = 1 - \frac{TS_{in}(1 - S_{out})(1/TS_{removed} - 1)}{1 - TS_{in}} \quad \text{(Eq. 10)}$$

where L_{out} is the liquid retention coefficient, TS_{in} is the incoming total solids fraction, S_{out} is the solids retention coefficient, and $TS_{removed}$ is the total solids fraction of the removed separated solids or stacked manure fraction.

A variable analysis was performed across a range of assumed influent manure TS values and separated cake TS values to evaluate how liquid retention changes across manure stream types (see Appendix A). This analysis was used to develop baseline recommendations for average U.S. conditions.

The variable analysis evaluated each separation technology across three representative influent manure solids-loading conditions: low-solids dilute flush manure at 1% TS, typical flush/lagoon manure at 4% TS, and higher-solids scrape/vacuum manure at 10% TS. The assumed separated cake solids contents were 15% TS for slope screen, 25% TS for screw press, 12% TS for rotary drum, 12% TS for vibratory/shaker screen, 25% TS for centrifuge, and 8% TS for settling basin. For each combination of influent TS, cake TS, and solids retention, liquid retention was calculated from the solids-liquid mass balance (Equation 10) to estimate how much liquid was entrained with the removed solids. The typical flush/lagoon case was used as the recommended baseline for most separation technologies because it represents a central condition for broad U.S. dairy screening. Screw press and centrifuge were split into low- and high-solids categories because their solids capture and liquid entrainment vary more strongly with influent TS [4]. For those two technologies, the low-solids dilute-flush case was used to define low-solids coefficients suitable for dilute or flush-based manure streams, while the high-solids scrape/vacuum case was used to define high-solids coefficients suitable for thicker manure streams.

The analysis confirmed that previous liquid coefficients for some fiber separation technologies were likely too aggressive if interpreted as bulk liquid retention. For example, a screw press liquid coefficient of 0.60 would imply that 40% of incoming liquid leaves with separated solids, which would require an extremely wet cake or very high solids capture. Under more realistic manure TS and cake TS assumptions, liquid retention for screens and screw presses generally remains closer to 0.90 to 0.99, while centrifugation can remove more liquid because of higher solids capture.

2.3 Storage Systems

Liquid retention in storage and digestion systems was simplified into two categories: open systems and covered systems. This categorization was selected because evaporation is the primary screening-level driver of liquid loss when flush water is assumed to cycle back to the lagoon or storage system.

Open systems include primary lagoons, secondary lagoons, effluent ponds, and uncovered liquid/slurry storage. These systems were assigned a liquid retention coefficient of 0.90. This value represents moderate net evaporative loss from exposed manure storage surfaces. EPA guidance supports accounting for evaporation and rainfall in lagoon volume design, but the 0.90 coefficient is a screening-level modeling assumption rather than a published EPA default.

Covered systems include covered lagoon digesters, plug-flow digesters, complete-mix digesters, fixed-film digesters, covered digestate storage, and liquid/slurry storage with a natural crust cover. These systems were assigned a liquid retention coefficient of 0.98. This value represents near-conservative hydraulic flow with minimal evaporative loss because the manure surface is covered or partially protected. The coefficient allows for minor residual vapor losses, condensate handling, operational losses, and imperfect hydraulic accounting.

All updated liquid retention coefficients are provided in Table 2.

Table 2. Updated screening-level liquid retention coefficients for dairy manure management systems.

Manure Management System	Liquids Change (Prev.)	Liquids Change (Updated)	Methods/Justification
Removal-Gutter / Chain and Paddle	0.95	0.95	Unchanged. Guided conveyance should retain most liquid manure, with only minor residual liquid left behind.
Removal-Scrape (Tractor / Skid)	0.9	0.7	Updated. Skid/tractor scraping captures bulk manure but is less effective at recovering free liquid and thin liquid films than guided conveyance or vacuum systems.
Removal-Scrape (Hydraulic / Mechanical)	0.85	0.75	Updated. Mechanical scraping improves collection consistency but still leaves more free liquid behind than vacuum, gutter, or flush systems.
Removal-Vacuum Truck	0.7	0.85	Updated. Vacuum collection should retain liquid more effectively than scrape systems because it is designed to recover liquid/slurry manure.
Removal-Flush (High Volume)	20	20	Unchanged. Treated as a dilution multiplier, not a liquid retention factor.
Removal-Flush (Automated/Precision)	8	8	Unchanged. Treated as a lower flush-water dilution multiplier than high-volume flush.
Removal-Gravity Flow / Slatted Floor	0.9	0.9	Unchanged. Gravity/slatted systems should retain most liquid in the downstream manure stream.
Storage-Primary Lagoon	0.8	0.9	Screening-level assumption representing moderate net evaporative loss from exposed lagoon or slurry storage surfaces. EPA supports adjusting lagoon volume for evaporation/rainfall, but the 0.90 coefficient is a modeling assumption, not an EPA default.
Storage-Secondary Lagoon/Effluent Pond	0.9	0.9	Screening-level assumption representing moderate net evaporative loss from exposed lagoon or slurry storage surfaces. EPA supports adjusting lagoon volume for evaporation/rainfall, but the 0.90 coefficient is a modeling assumption, not an EPA default.
Liquid/Slurry (Natural Cover) - Added	**-**	**0.98**	**Screening-level assumption representing near-conservative hydraulic flow with minimal evaporation because the manure surface is covered.**
Liquid/Slurry (Uncovered) - Added	**-**	**0.9**	**Screening-level assumption representing moderate net evaporative loss from exposed lagoon or slurry storage surfaces. EPA supports adjusting lagoon volume for evaporation/rainfall, but the 0.90 coefficient is a modeling assumption, not an EPA default.**
Storage-Digester (Covered lagoon)	0.98	0.98	Screening-level assumption representing near-conservative hydraulic flow with minimal evaporation because the manure surface is covered.
Storage-Digester (Plug-flow)	0.98	0.98	Screening-level assumption representing near-conservative hydraulic flow with minimal evaporation because the manure surface is covered.
Storage-Digester (Complete-mix)	0.98	0.98	Screening-level assumption representing near-conservative hydraulic flow with minimal evaporation because the manure surface is covered.
Storage-Digester(Fixed-film)	0.98	0.98	Screening-level assumption representing near-conservative hydraulic flow with minimal evaporation because the manure surface is covered.
Storage-Digester (Other/Specialized)	0.98	0.98	Screening-level assumption representing near-conservative hydraulic flow with minimal evaporation because the manure surface is covered.
Fiber Separation-Slope Screen	0.9	0.93	Based on solids removal and cake moisture – See Appendix A
Fiber Separation-Screw Press (High Solids)	0.6	0.83	Based on solids removal and cake moisture – See Appendix A
Fiber Separation-Screw Press (Low Solids) - Added	**-**	**0.99**	**Based on solids removal and cake moisture – See Appendix A**
Fiber Separation-Rotary Drum	0.75	0.92	Based on solids removal and cake moisture – See Appendix A
Fiber Separation-Vibratory / Shaker Screen	0.8	0.95	Based on solids removal and cake moisture – See Appendix A
Fiber Separation-Centrifuge (High Solids)	0.5	0.77	Based on solids removal and cake moisture – See Appendix A
Fiber Separation-Centrifuge (Low Solids) - Added	**-**	**0.98**	**Based on solids removal and cake moisture – See Appendix A**
Fiber Separation-Settling Basin	0.8	0.76	Based on solids removal and cake moisture – See Appendix A

Solids Removal-Stacking, Bedding, Etc.	0.85	0.85	Unchanged - assumes only a small fraction of liquid is removed with solids-removal
Field / Off-Site	1	1	Unchanged
Irrigation-Sand Pots / Sand Media Filters	1	1	Unchanged
N/A (No Treatment)	1	1	Unchanged

3. Nitrogen Retention Coefficients

3.1 Collection Systems

Nitrogen retention during collection was based on the barn/parlor nitrogen loss assumptions from Greene et al. [1], which were developed from the IPCC nitrogen emissions framework [2]. In this framework, EF_3 represents the fraction of managed manure nitrogen directly emitted as N_2O-N during the manure management stage, $Frac_{gas}$ represents the fraction of manure nitrogen volatilized as NH_3 and NO_x, and $Frac_{leach}$ represents the fraction of manure nitrogen lost through leaching or runoff. For the purpose of calculating the TN liquid multiplier, these factors were treated as nitrogen removed from the managed liquid manure stream. Indirect emission factors such as EF_4 and EF_5 were not subtracted from TN because they describe the fraction of volatilized or leached nitrogen later converted to N_2O, not additional nitrogen loss from the manure stream. Nitrogen retention was therefore calculated as the incoming nitrogen remaining after direct N_2O loss, volatilization, and leaching:

$$TN_{ret} = 1 - EF_3 - Frac_{gas} - Frac_{leach} \qquad \text{(Eq. 11)}$$

For barn/parlor collection systems:

$$TN_{ret,barn} = 1 - 0 - 0.155 - 0 = 0.845 \qquad \text{(Eq. 12)}$$

For non-flush collection systems, the barn/parlor nitrogen retention factor was applied directly as the TN liquid multiplier because no major dilution term was introduced during conveyance. For flush systems, the same retained nitrogen mass was divided by the flush liquid multiplier to represent the relative change in liquid-phase TN concentration after dilution. Therefore, the flush TN liquid multiplier reflects both nitrogen retention during barn/parlor handling and dilution from added flush water, not the absolute TN concentration of the effluent.

$$TN_{ret,high\ volume\ flush} = 0.845/20 = 0.042 \qquad \text{(Eq. 13)}$$

$$TN_{ret,precision\ flush} = 0.845/8 = 0.106 \qquad \text{(Eq. 14)}$$

3.2 Separation Systems

For mechanical fiber separation systems, no separate TN loss multiplier was applied. The model assigns nitrogen to the solids and liquid pools independently, so TN changes in the liquid effluent emerge from the retained solids-associated N and retained liquid-associated N after separation. Applying an additional empirical TN loss factor would double-count nitrogen removal.

The resulting modeled TN removal from fiber separation was checked against reported literature values for nitrogen recovery/removal in separated solids. The validation calculation used manure as excreted with 87% liquid and 13% solids, liquid-phase nitrogen concentration of 1%, and solids-phase nitrogen concentration of 2.5%. The equations were:

$$TN_{in} = (0.87 \times 0.01) + (0.13 \times 0.025) = 0.01195 \quad \text{(Eq. 15)}$$

$$TN_{out} = (0.87 \times 0.01 \times L_{ret}) + (0.13 \times 0.025 \times S_{ret}) \quad \text{(Eq. 16)}$$

$$TN_{removal,fiber\ separation} = 1 - \frac{TN_{out}}{TN_{in}} \quad \text{(Eq. 17)}$$

The resulting TN removal estimates are summarized in Table 3 and were used as a validation check for the updated fiber separation coefficients rather than as independent TN multipliers. These results show that the modeled nitrogen removal produced by solids and liquid partitioning generally falls within or near literature-reported ranges for mechanical manure separation, supporting the decision to set the TN liquid multiplier to 1.0 for mechanical fiber separation systems and allow nitrogen removal to emerge from the model's solids and liquid mass balance.

Table 3. Mechanistic TN removal check for fiber separation coefficients.

Separation technology	Solids Retention (S_{ret})	Liquids Retention (L_{ret})	Resulting TN Removal	Literature Range
Fiber Separation-Slope Screen	0.70	0.93	13.3%	13–24%
Fiber Separation-Screw Press (High Solids)	0.50	0.83	26.0%	13–24%
Fiber Separation-Screw Press (Low Solids)	0.75	0.99	7.5%	13–24%
Fiber Separation-Rotary Drum	0.75	0.92	12.6%	13–24%
Fiber Separation-Vibratory / Shaker Screen	0.85	0.95	7.7%	13–24%
Fiber Separation-Centrifuge (High Solids)	0.30	0.77	35.8%	13–24% for coarse fiber separation; higher removal expected for aggressive centrifugal separation
Fiber Separation-Centrifuge (Low Solids)	0.50	0.98	15.1%	13–24%
Fiber Separation-Settling Basin	0.50	0.76	31.1%	13–24% for coarse fiber separation; higher removal plausible for settling systems with substantial solids capture and residence time

The modeled TN removals were generally consistent with reported literature values for coarse fiber separation, which commonly fall near 13 to 24% TN recovery or removal in the separated solids fraction [7], [9], [10], [11], [12]. This supports treating TN removal during fiber separation as an emergent result of solids and liquid partitioning rather than applying a separate TN loss multiplier.

Settling basins were treated differently from mechanical fiber separators because they can include true nitrogen loss pathways during residence time [2]. For settling basins, the TN liquid multiplier was calculated using Equation 11 with nitrogen loss values from Greene et al. [1]:

$$TN_{ret,settling\ basin} = 1 - 0.005 - 0.300 - 0 = 0.695 \quad \text{(Eq. 18)}$$

This factor represents nitrogen retained after direct N_2O loss and volatilization, in addition to the physical solids and liquid partitioning captured by the solids and liquid coefficients.

3.3 Storage Systems

For storage systems, nitrogen retention coefficients were developed using the same nitrogen loss calculation described above and applied to the relevant storage-specific loss factors. Values were selected to reflect the dominant nitrogen loss pathways for each storage type, particularly ammonia volatilization from open liquid systems and reduced volatilization from covered or enclosed systems.

For liquid/slurry storage, lagoons, and effluent ponds, N loss factors were applied to the liquid-phase N pool. This is a simplifying allocation assumption rather than an explicit IPCC phase-specific treatment. The assumption is defensible because the dominant nitrogen loss pathway in these systems is ammonia volatilization, which occurs from aqueous TAN at the manure-air interface. Direct N2O and leaching losses were also applied to the liquid N pool for consistency, while solids-associated nitrogen was retained unless explicitly removed through physical separation or subsequent solids handling.

Ambient lagoons and effluent ponds were assigned:

$$TN_{ret,lagoon} = 1 - 0 - 0.35 - 0 = 0.65 \quad \text{(Eq. 19)}$$

Liquid/slurry with natural crust cover was assigned:

$$TN_{ret,slurry-covered} = 1 - 0.005 - 0.300 - 0 = 0.695 \quad \text{(Eq. 20)}$$

Uncovered liquid/slurry was assigned:

$$TN_{ret,slurry-uncovered} = 1 - 0 - 0.480 - 0 = 0.520 \quad \text{(Eq. 21)}$$

Covered lagoon digesters were assigned a TN multiplier of 0.995 when the row represents only the covered digester vessel. If the covered lagoon is followed by an effluent pond, the user should select secondary lagoon/effluent pond as the next downstream system to capture nitrogen losses from the uncovered pond. Engineered digesters were assigned:

$$TN_{ret,digester} = 1 - 0.0006 - 0 - 0 = 0.9994 \quad \text{(Eq. 22)}$$

This reflects the assumption that enclosed digesters have minimal direct nitrogen loss, with most digestate nitrogen losses occurring later in storage or land application.

All updated nitrogen retention coefficients are provided in Table 4.

Table 4. Updated screening-level TN liquid multipliers for dairy manure management systems.

Manure Management System	TN Change (Prev.)	TN Change (Updated)	Justification and References
Removal-Gutter / Chain and Paddle	1	0.845	Updated using Barn/Parlor N loss from Greene et al. 2024: (1 - 0 - 0.155 - 0 = 0.845)
Removal-Scrape (Tractor / Skid)	1	0.845	Updated using Barn/Parlor N loss from Greene et al, 2024: (1 - 0 - 0.155 - 0 = 0.845)
Removal-Scrape (Hydraulic / Mechanical)	1	0.845	Updated using Barn/Parlor N loss from Greene et al. 2024: (1 - 0 - 0.155 - 0 = 0.845)
Removal-Vacuum Truck	1	0.845	Updated using Barn/Parlor N loss from Greene et al. 2024: (1 - 0 - 0.155 - 0 = 0.845)
Removal-Flush (High Volume)	0.05	0.042	Updated using Barn/Parlor N loss from Greene et al. 2024: (1 - 0 - 0.155 - 0 = 0.845)

Removal-Flush (Automated/Precision)	0.125	0.106	Updated using Barn/Parlor N loss from Greene et al. 2024: (1 - 0 - 0.155 - 0 = 0.845)
Removal-Gravity Flow / Slatted Floor	1	0.845	Updated using Barn/Parlor N loss from Greene et al. 2024: (1 - 0 - 0.155 - 0 = 0.845)
Storage-Primary Lagoon	0.8	0.65	Updated using Ambient Lagoon N loss from Greene et al. 2024: (1 - 0 - 0.35 - 0 = 0.65)
Storage-Secondary Lagoon/Effluent Pond	0.9	0.65	Updated using Ambient Lagoon N loss from Greene et al. 2024: (1 - 0 - 0.35 - 0 = 0.65)
Liquid/Slurry (Natural Cover) - Added	**-**	**0.695**	**Updated using IPCC-based N loss from Greene et al. 2024: (1−0.005−0.300−0=0.6951)**
Liquid/Slurry (Uncovered) - Added	**-**	**0.52**	**Updated using IPCC-based N loss from Greene et al. 2024: (1−0−0.480−0=0.520)**
Storage-Digester (Covered lagoon)	0.995	0.995	Unchanged - If the row represents only the covered digester vessel, keep 0.995. If effluent pond is used, user should select secondary lagoon after lagoon digester.
Storage-Digester (Plug-flow)	1	0.9994	Assumed same as complete mix.
Storage-Digester (Complete-mix)	1	0.9994	Updated using CMAD direct N2O loss from Greene et al. 2024: (1 - 0.0006 - 0 - 0 = 0.9994).
Storage-Digester (Fixed-film)	1	0.9994	Assumed same as complete mix.
Storage-Digester (Other/Specialized)	1	0.9994	Assumed same as complete mix.
Fiber Separation-Slope Screen	1	1	Captured in solid/liquid retention coefficients - see Table 3.
Fiber Separation-Screw Press (High Solids)	1	1	Captured in solid/liquid retention coefficients - see Table 3.
Fiber Separation-Screw Press (Low Solids) - Added	**-**	**1**	**Captured in solid/liquid retention coefficients - see Table 3.**
Fiber Separation-Rotary Drum	1	1	Captured in solid/liquid retention coefficients - see Table 3.
Fiber Separation-Vibratory / Shaker Screen	1	1	Captured in solid/liquid retention coefficients - see Table 3.
Fiber Separation-Centrifuge (High Solids)	1	1	Captured in solid/liquid retention coefficients - see Table 3.
Fiber Separation-Centrifuge (Low Solids) - Added	**-**	**1**	**Captured in solid/liquid retention coefficients - see Table 3.**
Fiber Separation-Settling Basin	1	0.695	Updated using Settling Basin N loss from Greene et al. 2024: (1 - 0.005 - 0.30 - 0 = 0.695)
Solids Removal-Stacking, Bedding, Etc.	1	1	Solids partitioning, no physical loss pathways for N
Field / Off-Site	1	1	Unchanged
Irrigation-Sand Pots / Sand Media Filters	1	1	Unchanged
N/A (No Treatment)	1	1	Unchanged

Interpretation and Recommendations

The updated coefficients provide a more transparent and internally consistent framework than the prior placeholder set. The most important conceptual improvement is that coefficients now represent different physical mechanisms depending on subsystem type. Collection systems are treated as transfer efficiencies. Fiber separation systems are treated as physical partitioning systems. Storage and digestion systems are treated as biological conversion and liquid-loss systems. Nitrogen retention is treated as a function of IPCC-style direct N_2O, volatilization, and leaching loss pathways, except for mechanical separation systems where nitrogen partitioning is already captured through the independent solids and liquid pools.

For fiber separation, the updated methodology avoids applying an empirical TN loss factor on top of solids and liquid partitioning. This is important because the model already assigns nitrogen to solids and liquid fractions. If a separate TN loss coefficient were applied to mechanical separators, the model would double-count nitrogen removal. The validation check showed that the resulting TN removals are broadly consistent with reported literature values, particularly for slope screens, rotary drums, low-solids centrifuges, and typical coarse-fiber separation. Higher-removal cases such as high-solids screw press, high-solids centrifuge, and settling basin systems exceed the coarse-fiber literature range, but this is plausible because those systems either remove more solids, retain more liquid in the separated solids fraction, or include true nitrogen loss pathways, respectively.

The recommendation to split screw press and centrifuge systems into high-solids and low-solids categories strengthens the model. These technologies can behave very differently depending on influent manure TS. A single coefficient cannot represent both dilute flush manure and higher-solids scrape or vacuum manure. The added categories allow the user to select coefficients that better match the manure stream being treated.

For storage systems, the updated TS retention method improves consistency by converting VS-based degradation assumptions to TS retention using appropriate VS/TS ratios. This prevents overestimating TS destruction by applying MCF values directly to total solids. Lagoon systems use a lower VS/TS ratio because lagoon water contains a larger fixed-solids fraction and reflects dilution, settling, inorganic accumulation, and biological degradation. Liquid/slurry systems use a higher VS/TS ratio because scrape/vacuum manure is less dilute and less stabilized.

The addition of liquid/slurry with natural cover and liquid/slurry uncovered is strongly recommended. Liquid/slurry storage is a common standalone manure management system and also serves as a digestate storage pathway after engineered digestion. The natural-cover and uncovered categories allow the model to account for differences in both methane conversion and nitrogen volatilization. This also improves pathway construction. For example, after a covered lagoon digester, the user can select secondary lagoon/effluent pond to represent downstream effluent storage. After engineered digestion, the user can select liquid/slurry with natural cover or uncovered liquid/slurry to represent digestate storage prior to land application.

The liquid retention coefficients remain screening-level assumptions. A detailed liquid balance would require storage geometry, climate, surface area, residence time, precipitation, evaporation, runoff, seepage, and recycling of flush water. Because these data are not available for screening-level modeling, the simplified covered/open approach is appropriate.

The nitrogen retention coefficients are more defensible than the prior placeholder values because they are tied to direct N_2O, volatilization, and leaching fractions. For lagoons and liquid/slurry systems, ammonia volatilization is the dominant N loss pathway, so applying N loss to the liquid-phase N pool is a reasonable model allocation assumption. The model should still document that IPCC does not explicitly assign N losses to liquid or solids phases, but rather assigns losses at the manure management system level [2]. The liquid-phase allocation is a simplifying assumption based on the dominant role of aqueous TAN and ammonia volatilization at the manure-air interface.

The solids removal/stacking row should be interpreted carefully. Under the current coefficients, the row implies removal of 90% of incoming solids and 15% of incoming liquid. For manure entering at 13% TS, this implies the removed stacked manure fraction is approximately 47% TS:

$$TS_{stacked} = \frac{0.13(1-0.10)}{0.13(1-0.10)+0.87(1-0.85)} = 0.47 \quad \text{(Eq. 23)}$$

This is relatively dry but plausible if the row represents stackable or bedding-rich manure rather than fresh slurry. The row should be labeled and documented as manure stacking or stackable solids diversion rather than generic solids removal.

Conclusion

The updated coefficient set provides a transparent, repeatable, and defensible basis for screening-level modeling of dairy manure management systems. Collection coefficients were interpreted as physical transfer efficiencies. Fiber separation coefficients were based on solids removal, cake moisture, and liquid entrainment, with nitrogen changes captured mechanistically through solids and liquid partitioning. Storage and digestion solids coefficients were derived from VS destruction or MCF-based assumptions converted to TS retention using scenario-specific VS/TS ratios. Liquid coefficients were simplified into covered and open storage categories based on evaporation potential. Nitrogen retention coefficients were calculated using IPCC-style direct N_2O, volatilization, and leaching loss factors, with values from Greene et al. [1] used to fill gaps.

The most important recommended model updates are:

- Add low- and high-solids categories for screw press and centrifuge separation
- Add liquid/slurry storage with natural cover and uncovered liquid/slurry storage
- Treat secondary lagoon as an effluent pond option after covered lagoon digestion
- Treat engineered digester and digestate storage as separate systems.

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Appendix A: Analysis of the Impact of Slurry Solids Content on Liquid Retention Coefficients.

Table A1. Variable analysis for fiber separation solids and liquid retention coefficients across representative manure solids-loading scenarios. Solids retention coefficients were based on reported or assumed solids removal performance, and liquid retention coefficients were calculated from the solids-liquid mass balance using influent TS, solids retention, and separated cake TS. The selected baseline recommendation for each technology is highlighted in green (LS = low solids, HS = high solids).

Assumed Cake % Solids	Coefficient →	Solids Retention Coefficient			Liquid Retention Coefficient		
	Slurry Scenario →	Low Solids (Dilute Flush)	Typical Flush/Lagoon	Scrape/ Vacuum	Low Solids (Dilute Flush)	Typical Flush/Lagoon	Scrape/ Vacuum
	Manure Stream % Solids →	1%	4%	10%	1%	4%	10%
15%	Fiber Separation-Slope Screen	0.7	0.7	0.7	0.98	0.93	0.81
25%	Fiber Separation-Screw Press	0.75 (LS)	0.6	0.5 (HS)	0.99 (LS)	0.95	0.83 (HS)
12%	Fiber Separation-Rotary Drum	0.75	0.75	0.75	0.98	0.92	0.80
12%	Fiber Separation-Vibratory / Shaker Screen	0.85	0.85	0.85	0.99	0.95	0.88
25%	Fiber Separation-Centrifuge	0.5 (LS)	0.4	0.3 (HS)	0.98 (LS)	0.93	0.77 (HS)
8%	Fiber Separation-Settling Basin	0.5	0.5	0.5	0.94	0.76	0.36*

*settling basin not usually used with scrape systems

Table A2. Recommended baseline coefficients for fiber separation systems.

Separation Technology	Recommended Baseline Coefficients		
	Solids Retention	**Liquid Retention**	**TN Liquid Coefficient**
Fiber Separation-Slope Screen	0.7	0.93	1
Fiber Separation-Screw Press (High Solids)	0.5	0.83	1
Fiber Separation-Screw Press (Low Solids)	0.75	0.99	1
Fiber Separation-Rotary Drum	0.75	0.92	1
Fiber Separation-Vibratory / Shaker Screen	0.85	0.95	1
Fiber Separation-Centrifuge (High Solids)	0.3	0.77	1
Fiber Separation-Centrifuge (Low Solids)	0.5	0.98	1
Fiber Separation-Settling Basin	0.5	0.76	1